REFERENCE TRANSLATION prepared by Van Phong Technical and Trading Co., Ltd. The published original is the authoritative text; see it for figures.

Effect of Cold Treatment on the Wear Resistance of Hydraulic Breaker Picks

Le Thanh Binh, V.I. Bolobov

Tạp chí Công Thương (Industry and Trade Magazine) – Science & Technology Special Issue (Chuyên san Khoa học & Công nghệ), ISSN 0866-7756, No. 43 (10/2020), pp. 12–14, 18

Abstract

The paper presents the results of a study determining the wear intensity of samples of 38XM, Y8 and X12M steels, which are commonly used to manufacture hydraulic breaker picks, with an additional cold treatment step in the heat treatment process. Cold treatment improves the strength and wear resistance of all three steels, while not significantly affecting their impact toughness. Comparison of wear intensity and hardness values, together with analysis of the microstructure of the steel samples, leads to the conclusion that wear intensity depends not only on hardness but also on the type, quantity, shape and distribution of the carbides formed in the steel structure. The effect of cryogenic (deep cold) treatment on wear resistance is strongest for X12M steel (an increase of 98.9%), because this material has the highest carbon content (~1.6%) and alloying element content (~14%), which increases the amount of fine carbide particles in its structure by 10.8%.

Keywords: *carbide, wear intensity, granite, hydraulic breaker pick, cold treatment.*

1. Introduction

Breaking rock from the massif is a very complex process that requires considerable resources in terms of energy, equipment, materials and labour. Reducing resource consumption has always been a challenge and one of the important goals in the mining sector. Traditionally, rock-breaking methods are divided into the following groups: mechanical, blasting, physical, chemical and combined. Although there are many options among rock-breaking techniques, mechanical techniques remain the most widely used because of the low level of knowledge about alternative mining techniques and their high application costs. According to [1], mechanical techniques account for up to 85% of the total volume of excavation and mining work. Among mechanical techniques, hydraulic breakers are very widely used in many fields, such as: driving mine roadways and road tunnels through mountains; mineral extraction; breaking oversize rock; demolition of building and transport structures, etc.

During operation, high-intensity impacts between the pick and the rock accelerate its wear. For this reason, the pick must be made of a material with sufficient hardness, strength and good wear resistance. Usually, medium- and high-carbon steels with various alloy contents are used for its manufacture, combined with conventional heat treatment (CHT), comprising austenitization, quenching and low-temperature tempering. The structure of the steel after CHT is martensite with a certain amount of retained austenite and carbides. Retained austenite exists because the austenite–martensite transformation finish temperature M_f is lower than the temperature of the quenching

medium; it depends on the carbon and alloy content of the steel. The presence of "soft" austenite reduces the hardness, strength and wear resistance of the pick. Moreover, during operation, part of the retained austenite may transform into martensite. Since the volume of martensite is 4.3% greater than that of austenite, this transformation leads to dimensional distortion of the pick, affecting the motion and service life of the hydraulic breaker. To reduce retained austenite and increase the amount of carbides in steels, many studies on cold treatment have been carried out to increase the hardness and wear resistance of steels. This paper investigates the effect of cold treatment on the structure, mechanical properties and wear resistance of 38XM, Y8 and X12M steels, which are commonly used to manufacture hydraulic breaker picks.

2. Experimental

Test samples made of the three steels 38XM, Y8 and X12M were oil-quenched from 800–1000 °C depending on the steel grade and tempered at a low temperature of 200 °C. Samples with an additional sub-zero cold treatment step (SCT) were placed in a refrigerating cabinet at −75 °C for 5 hours immediately after quenching and before low-temperature tempering. For the cryogenic (deep cold) treated samples (DCT), the cooling medium was liquid nitrogen (−196 °C) held in a special vessel, with a holding time of 72 hours. The general heat treatment scheme is shown in Figure 1.

[Figure 1 — see original PDF, page 2] Heat treatment scheme

Vickers hardness HV was measured with a Wilson Wolpert 751N hardness tester in accordance with TCVN 258-1:2007. Impact toughness KCV specimens had standard dimensions and were tested on a JBS-300 impact tester at room temperature (~20 °C) in accordance with TCVN 312-1:2007. Microstructure images were taken with an Axioplan 2 optical microscope combined with the accompanying dedicated calculation software, using colour etching with an NaOH + K_3FeN_6 etchant to determine the amount of carbides, and an HCl + picric acid etchant to determine the total content of retained austenite and carbides. Measurements were averaged over 5 locations on each sample to limit error. From this, the retained austenite content was determined:

$$\%\ retained\ austenite = \%(retained\ austenite + carbides) - \%\ carbides$$

The wear intensity test was carried out on an apparatus shown schematically in Figure 2, with a main spindle rotation frequency n = 7.5 s^{-1}. The abrasive material 1 was a granite block — a material commonly broken by hydraulic breakers. As the spindle rotates, the end face of steel sample 2, 8 mm in diameter, slides over the surface of the rock block under a constant static load P = 100 N, producing a wear track on the block about 8 mm wide with a radius to the track centreline R = 18 mm. The average sliding speed of the steel sample is then v = 2πRn = 0.85 m/s. The sliding time was t = 50 min, and the sample was weighed every 10 minutes with a measurement accuracy of 1 mg to calculate the worn mass Δm. During sliding, the sample was cooled with water.

[Figure 2 — see original PDF, page 3] Schematic of the wear intensity test apparatus: 1 — granite block; 2 — steel sample; 3 — sample holder; 4 — main spindle of the apparatus; 5 — water supply device

3. Results and Discussion

From the test results, plots of the dependence of the total worn mass ΣΔm on the sliding distance over the granite block L were constructed, where L = v·t. The dependence of the total worn mass ΣΔm on the sliding distance L for the steel samples is shown in Figure 3.

[Figure 3 — see original PDF, page 3] Dependence of the total worn mass on the sliding distance over the granite block for samples of 38XM (a), Y8 (b) and X12M (c) steels with different heat treatment methods: 1 — CHT; 2 — SCT; 3 — DCT

The test results show that, for all steel samples, their wear on granite from the very beginning of contact can be described by a linear dependence $\Sigma\Delta m = I \cdot L$ with a coefficient of determination R^2 ~0.99. The slope (tangent of the inclination angle) of the $\Sigma\Delta m$ lines is the wear intensity I [mg/m], which differs for each steel and heat treatment method. The reciprocal of I is taken as the wear resistance W_R [m/mg] of the steel under these abrasive conditions.

Comparison of the slopes of the $\Sigma\Delta m$ lines in Figure 3 leads to the conclusion that the X12M samples, under all heat treatment methods, have the highest wear resistance, while the 38XM samples have the lowest. Both sub-zero cold treatment SCT and cryogenic treatment DCT increase the wear resistance of all three steels.

The specific values of wear resistance W_R, hardness HV and impact toughness KCV of the steel samples are given in Table 1. Figures 4–5–6 show the dependence of these properties on the cold treatment temperature.

Table 1 – Test results for the physical-mechanical properties and wear resistance of the steel samples under different heat treatment regimes

Steel grade	Heat treatment method	Measured values		
		KCV, J/cm^2	HV	I, mg/m (W_R, m/mg)
38XM	CHT	40.6	529	0.040 (25.2)
	Quenching + SCT + tempering	33.0	536	0.032 (31.1)
	Quenching + DCT + tempering	35.0	539	0.030 (33.9)
Y8	CHT	5.1	707	0.031 (31.9)
	Quenching + SCT + tempering	3.9	717	0.029 (34.8)
	Quenching + DCT + tempering	5.1	730	0.026 (38.6)
X12M	CHT	3.1	694	0.018 (57.1)
	Quenching + SCT + tempering	2.5	708	0.013 (75.2)
	Quenching + DCT + tempering	4.1	750	0.009 (113.6)

[Figure 4 — see original PDF, page 4] Effect of cryogenic treatment temperature on the hardness of the materials: 1 — 38XM steel; 2 — Y8 steel; 3 — X12M steel

[Figure 5 — see original PDF, page 4] Effect of cryogenic treatment temperature on the impact toughness of the materials: 1 — 38XM steel; 2 — Y8 steel; 3 — X12M steel

[Figure 6 — see original PDF, page 4] Effect of cryogenic treatment temperature on the wear intensity of the materials: 1 — 38XM steel; 2 — Y8 steel; 3 — X12M steel

The results obtained show that cold treatment of the steel samples, along with improving hardness HV, also increases their wear resistance W_R. This effect is particularly pronounced for X12M steel after cryogenic treatment at −196 °C: W_R increases by 98.9% and HV by 8.2%. The impact

toughness of the steel samples decreases slightly after cold treatment at −75 °C; however, if the treatment temperature is lowered to −196 °C, impact toughness recovers and even equals or exceeds the value after conventional heat treatment for Y8 and X12M steels. This is a very important feature and can be explained by the results of the microstructural study of the steel samples presented in Table 2.

Table 2 – Retained austenite and carbide contents in the steel samples depending on the cold treatment regime

Steel grade	% retained austenite		
	CHT	SCT	DCT
38XM	≤0.10	≤0.10	≤0.10
Y8	2.21	1.24	0.84
X12M	15.65	2.78	0.50
Steel grade	**% carbides**		
	CHT	**SCT**	**DCT**
38XM	0.17	0.33	0.58
Y8	≤0.10	≤0.10	≤0.10
X12M	5.41	6.80	10.84

The data in Table 2 show that cold treatment of the high-carbon steel samples Y8 (~0.8% C) and X12M (~1.6% C) leads to a significant reduction in the content of soft retained austenite, i.e. an increase in the content of hard martensite in the structure. It also helps increase the amount of carbides in the alloyed steel samples 38XM (~2%) and X12M (~14%). Moreover, the lower the cold treatment temperature, the greater these effects. It can be concluded that the wear resistance of steel depends not only on its hardness but also on the type, quantity, shape and distribution of carbides in the structure. The X12M sample after DCT has a carbide content in its structure of up to 10.8%, since it contains the highest carbon (~1.6%) and alloying element (~14%) contents, and shows the best wear resistance. The improved impact toughness of the steel samples after DCT can be explained by the uniform distribution of homogeneous, fine carbides in the structure [2–4]. In addition, it may partly be because the precipitated carbides reduce a certain amount of supersaturated carbon in the martensite crystals, leading to reduced brittleness.

4. Conclusions

From the results presented, the following conclusions can be drawn:

1) Cold treatment can be regarded as a promising additional operation in the heat treatment of hydraulic breaker picks to increase their wear resistance. The preferred treatment temperature should be −196 °C, at which, along with increased hardness and better wear resistance, the impact toughness of the steel is improved owing to the uniform distribution of homogeneous and fine carbides in the structure, and to the precipitated carbides reducing a certain amount of supersaturated carbon in the martensite crystals.

2) Among the 3 materials studied, the effect of cryogenic treatment on wear resistance is strongest for X12M steel, because it has the highest carbon and alloying element contents.

3) The wear resistance of steel depends not only on its hardness but also very strongly on the type, quantity, shape and distribution of carbides in the structure.

References

1. Бритарев В.А., Замышляев В.Ф. Горные машины и комплексы. М.: Недра. 1984. 288 с..
2. Bensely A., Venkateswaran S., Subisak A.D., Lal D.M., Rajadurai A., Lenkey G.B., Paulin P. Influence of Deep Cryogenic Treatment on Alloy Carbide Precipitations and Mechanical Properties of AISI M2 High Speed Tool Steel. Cold Facts. 2012. 28(2). PP. 4-9.
3. Das D., Sarkar R., Dutta A.K., Ray K.K. Influence of Sub-zero Treatments on Fracture Toughness of AISI D2 Steel. Materials Science and Engineering A. 2010. Vol. 528. PP. 589-603.
4. Sonara T., Lomtea S.V., Gogteb C.L. Cryogenic Treatment of Metal – A Review. Materials Today: Proceedings. 2018. Vol. 5(11). PP. 25219-25228.

Received: 12/9/2020; reviewed and revised: 22/9/2020; accepted for publication: 27/9/2020.